

LLMs Out-of-the Box Do **Not** Generate Context-Appropriate Word Order in Russian

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Word Order Variation in Russian

- | | |
|---|-----|
| 1. Maria čitaet knigu.
Maria.NOM read.PRES.3SG book.ACC
<i>‘Maria reads a book’</i> | SVO |
| 2. Maria knigu čitaet. | SOV |
| 3. Knigu Maria čitaet. | OSV |
| 4. Knigu čitaet Maria. | OVS |
| 5. Čitaet Maria knigu. | VSO |
| 6. Čitaet knigu Maria. | VOS |

Examples

- Нет, я работаю в кино.
- Я и режиссер, и актер, и сценарист и продюсер.

- Вы меня дурачите!

SOV

- Ты рассказала его пятидесяти миллионам зрителей, а мне не хочешь?
- Обещаю, я не буду смеяться.

- Этого-то я и опасаясь.

OSV

- No, I work in film.
- I am a director, an actor, a screenwriter, and a producer.

- You're fooling me!

- You told it to fifty million viewers, but you won't tell me?
- I promise I won't laugh.

- That is exactly what I fear.

Research Questions

1. What **features** influence surface word order in Russian?
2. Can **LLMs** generate various word orders in Russian, or do they default to the canonical SVO order?

Outline

1. Corpus

2. Machine Learning Experiments

3. LLM Experiments

Corpus & Data Preprocessing

- Based on the Movie Scripts dataset (Lisovsky, 2019);
- 20 film scripts translated into Russian;
- Genres: crime, comedy, drama, thriller (1992-2012);
- 10 scripts manually edited so that it can be used as a dialogue corpus;
 - We removed screenplay elements such as camera instructions, scene descriptions and speaker names.

Building the Corpus

1. Word order information based on dependency relations (Stanza)

Вопросы тут задаю я.

OVS

I'm the one asking the questions here.

2. Preceding context

- Это не твой текст. Скажи,
что любишь меня.

- It's not your text. Tell me
you love me.

- Я тебя люблю.

SOV

- I love you.

Word Order Distribution

Word order	Count	Percentage	Binary
SVO	1123	72.8%	1123 (72.8%)
SOV	275	17.8%	420 (27.2%)
OSV	71	4.6%	
OVS	50	3.2%	
VSO	14	0.9%	
VOS	10	0.7%	

Outline

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Decision Tree: binary classification

Class	Precision	Recall	F1
SVO (74%)	0.88	0.83	0.86
non-SVO (26%)	0.59	0.68	0.63

Metric	Accuracy on Test set	Majority Baseline (SVO)	Random Baseline
Accuracy	0.79	0.43	0.50

Feature	Imp.
object_pos	0.61
predicate_type	0.10
coref_score_object	0.06
has_negation	0.05
verb_constituent_size	0.04
object_syllable_weight	0.03
object_ner	0.02
subject_role	0.02
focus_subject	0.02
verb_type	0.01

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Prompt Template

You are working with a movie script in Russian. Below is a context from the script, where:

SPEECH indicates character dialogue

COMMENT indicates scene description or comment

Context:

{context}

Continue this movie scene by generating a new sentence that naturally follows this context.

The generated sentence **MUST** contain:

- the verb "{verb_lemma}" (with the specified aspect, tense, and negation, if applicable),
- "{subject}" as the subject constituent (nominative case), and
- "{object}" as the object constituent (accusative case).

Requirements:

1. You must use the exact given words:
 - subject: "{subject}"
 - verb: "{verb_lemma}"
 - object: "{object}"
2. These words must be used within the same clause as direct arguments of the verb.
3. Generate exactly one sentence in Russian. Do not add translation, comment, or explanation.
4. The sentence must:
 - be in the indicative mood,
 - be natural continuation of the context,
 - be grammatically correct,
 - not be a question, command, or use the subjunctive mood,
 - no be a copy of the context.

Prompt continued

NOTE ON WORD ORDER:

While Russian allows flexible word order, subject-verb-object (SVO) is the most common and natural patterns. Prefer SVO unless the context (pragmatic or stylistic features or morphosyntactic properties) require a different order for emphasis, information structure, style, or emotion.

OUTPUT:

Provide only one generated Russian sentence.

Examples

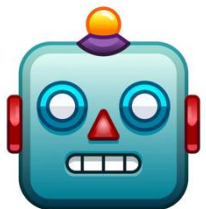
- Ты хотел собой покончить.
- А ты мне пытался мозги вышибить.

- You wanted to kill yourself.
- And you tried to blow my brains out.

- **Пистолет** ты не получишь.

OSV

- You won't get a gun.



1. Ты не получил **пистолет**, чтобы всё это закончить.

2. Ты не смог получить **пистолет**.

3. Ты не получил **пистолет**, чтобы сдаться.

SVO

1. You didn't get a gun to end it all.

2. You couldn't get a gun.

3. You didn't get a gun to give up.

Model Selection

Goal: **NOT** to evaluate whether the model generates the “correct” word but investigate **word order distribution**.

- Models evaluated: GPT-3.5 Turbo, GPT-4o-mini, and GPT-5-mini
- 3 runs per sentence

	Gold	GPT-3.5-turbo	GPT-4o-mini	GPT-4o
SVO	74.0%	83.6%	90.1%	86.4%
other	26.0%	7.0%	0.5 %	5.9%
UNK		9.4%	9.3%	7.7%

Unknown (UNK) outputs generated sentences that could not be evaluated due to:

- Incoherent output;
- Failure to follow the instructions in the prompt

Include Decision Tree Prediction in Prompt

WO	GOLD	GPT-4omini	Decision Tree predictions in Prompt
SVO	74.0%	90.1%	85.6%
other	26.0%	0.5%	11.6%
UNK		9.3%	2.7%
accuracy		69.3%	73.4%

Unknown (UNK) outputs

Generated sentences that could not be evaluated due to:

- Incoherent output;
- Failure to follow the instructions in the prompt

Few-shot prompting

Original prompt with four examples, one for each word order (with context):

SVO

SOV

OVS

OSV

Fine-tuning

- Supervised fine-tuning on the training corpus;
- Training examples formatted as input-output pairs;
- Input: context + constituent + morphosyntactic features from the prompt;
- Output: target word order label

Few-shot prompting & Fine-tuning

WO	GOLD	GPT-4omini	Decision Tree predictions in Prompt	Few-shot	Fine-tuned
SVO	74.0%	90.1%	85.6%	95.5%	71.7%
other	26.0%	0.5%	11.6%	0.8%	18.5%
UNK		9.3%	2.7%	3.6%	9.8%
accuracy		69.3%	73.4%	71.8%	73.6%

*Word order distributions across baseline annotations and multiple prompting strategies (**GPT-4o-mini**)*

Summary: Contributions

- Releasing an annotated corpus;
- Evaluation of machine learning experiments evaluating our set of features predicting word order;
- LLMs struggle to generate various word orders;
- Possible solution: fine-tuning.

THANK YOU!

BACKUP SLIDES

Data Annotation

Manually annotated 10 scripts (for script structure) using INCEpTION:

- COMMENT
- SPEECH
- CHARACTER NAME
- CAMERA

Movie Scripts (Lisovsky, 2019)

The screenshot displays the INCEpTION web interface for annotating a movie script. The top bar shows the file name 'pulpfiction.txt' and the current position '371-380 / 3377 sentences [doc 1 / 1]'. Below the bar is a toolbar with icons for file operations and navigation. The main area shows a list of script lines with manual annotations in colored boxes:

- Line 371: **(CHA...)** ДЖУЛС **(SPEECH)** Если любишь гамбургеры, попробуй вот эти как-нибудь.
- Line 372: **(SPEECH)** Я-то сам обычно их не ем, потому что моя девушка - вегетарианка.
- Line 373: **(SPEECH)** Из-за этого и я почти что вегетарианец, но со вкусом хорошего бургера ничего не сравнится.
- Line 374: **(CAMERA)** (Бретту) **(SPEECH)** Ты знаешь, как во Франции называют Четвертьфунтовик с Сыром?
- Line 375: **(CHA...)** БРЕТТ **(SPEECH)** Нет.
- Line 376: **(CHA...)** ДЖУЛС **(SPEECH)** Скажи ему, Винсент.
- Line 377: **(CHA...)** ВИНСЕНТ **(SPEECH)** Роял с Сыром.
- Line 378: **(CHA...)** ДЖУЛС **(SPEECH)** Роял с Сыром.
- Line 379: **(SPEECH)** А почему, знаешь?
- Line 380: **(CHA...)** БРЕТТ **(SPEECH)** Из-за метрической системы?

Building the Corpus: Adding Features

Morphosyntactic

- Part of speech (POS): noun, pronoun, verb, etc.;
- Constituent size (number of tokens) for subject, object, verb;
- Verb aspect (perfective / imperfective);
- Presence of negation for the verb;
- Clause type (matrix / embedded clause)

Semantic

- Subject and object named entity types (PER, ORG, LOC, MISC)
- Animacy for subject and object;
- Verb type (perception, communication, action, etc.)
- Semantic roles of subject and object (Experiencer, Agent, Theme, etc.)

Phonological

- Syllable weight

Discourse

- Discourse-related elements;
- The coreference score;
- Is target sentence preceded by a question?

Example

Ona ne polozhila tvoj dvojnoj burger!
She.NOM not put in. PAST.FEM. your double burger.ACC.
'She didn't put your double burger in.'

Ona ne polozhila tvoj dvojnoj burger

- **POS:** PRONOUN
- **constituent size:** 1
- **syllable weight:** 2
- **animacy:**
not_applicable
- **NER:** PER
- **type:** agent
- **coreference score:**
1.2

- **syllable weight:** 4
- **aspect:** perfective;
- **verb type:** action;
- **predicate type:** 2-
place

- **POS:** NOUN
- **constituent size:** 3
- **syllable weight:** 5
- **animacy:** inanimate
- **NER:** not_applicable
- **type:** theme
- **coreference score:**
1.2

- **preceded by**
question: FALSE;
- **presence of**
negation: TRUE;
- **clause type:** matrix;
- **WORD ORDER:** SVO
- **focus subject:** FALSE
- **focus object:** FALSE

Decision Tree: multiclass classification

class	precision	recall	F-1 score
SVO (74%)	0.88	0.85	0.87
SOV (18%)	0.53	0.78	0.63
OSV (5%)	0.75	0.21	0.33
OVS (3%)	0.33	0.10	0.15

Metric	Accuracy on Test set	Majority Baseline (SVO)	Random Baseline
accuracy	0.78	0.21	0.28

Top-10 most important features

Feature	Imp.
object_pos	0.47
coref_score_object	0.09
obj_animacy	0.09
predicate_type	0.08
object_syllable_weight	0.05
verb_constituent_size	0.05
subject_role	0.04
coref_score_subject	0.03
object_ner	0.03
subject_syllable_weight	0.02

Multiclass classification

Feature	Imp.
object_pos	0.61
predicate_type	0.10
coref_score_object	0.06
has_negation	0.05
verb_constituent_size	0.04
object_syllable_weight	0.03
object_ner	0.02
subject_role	0.02
focus_subject	0.02
verb_type	0.01

Binary classification